

Date

Date

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2402013

week 6

tutorial

1) Consider the following input images

$$I = \begin{bmatrix} 10 & 35 & 35 & 35 & 35 & 20 \\ 29 & 46 & 44 & 42 & 45 & 27 \\ 16 & 25 & 21 & 19 & 19 & 12 \\ 66 & 120 & 116 & 154 & 114 & 62 \\ 74 & 216 & 174 & 252 & 172 & 112 \\ 70 & 210 & 170 & 25 & 170 & 110 \end{bmatrix}$$

$$\text{Filter } K = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Stride = 1 no padding

Step 1

$$\text{Output size} = \frac{N-F}{S} + 1$$

$$N = 6$$

$$F = 3$$

$$S = 1$$

$$\text{So, } \frac{6-3}{1} + 1 = \frac{3}{1} + 1 = 4$$

So,

Convolution output will be
4x4

Since,

center of kernel is 0 we add
8 surrounding values,

first row

C_{11}

Region

$$\begin{bmatrix} 20 & 35 & 35 \\ 29 & 46 & 44 \\ 16 & 25 & 21 \end{bmatrix}$$

$$= 20 \times 1 + 35 \times 1 + 35 \times 1 + 29 \times 1 + 46 \times 1 + 44 \times 1 + 16 \times 1 + 25 \times 1 + 21 \times 1$$

$$= 255$$

$$C_{12} = \begin{bmatrix} 35 & 35 & 35 \\ 46 & 44 & 42 \\ 25 & 21 & 19 \end{bmatrix}$$

$$= 35 + 35 + 35 + 46 + 44 + 42 + 25 + 21 + 19$$

$$= 258$$

$$C_{13} = \begin{bmatrix} 35 & 35 & 35 \\ 44 & 42 & 42 \\ 21 & 19 & 19 \end{bmatrix}$$

$$= 35 + 35 + 35 + 44 + 42 + 42 + 21 + 19 + 19$$

$$= 250$$

$$C_{14} = \begin{bmatrix} 35 \\ 42 \\ 19 \end{bmatrix}$$

$$= 35 + 35 + 42 + 19$$

$$= 458$$

Second Row

$$C_{21} = \begin{bmatrix} 29 \\ 16 \\ 66 \end{bmatrix}$$

$$= 29 + 16 + 66$$

$$= 111$$

$$C_{22} = \begin{bmatrix} 46 \\ 25 \\ 120 \end{bmatrix}$$

$$= 46 + 25 + 120$$

$$= 191$$

$$= 566$$

$$C_{23} = \begin{bmatrix} 44 \\ 21 \\ 116 \end{bmatrix}$$

$$= 44 + 21 + 116$$

$$= 181$$

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0 we add

$$\begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$+ 23 \times 1 + 46 \times 0$$

$$\times 1 + 21 \times 1$$

$$12 + 25 + 21 + 9$$

$$- 42 + 21 + 19 + 9$$

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$$C_{14} = \begin{bmatrix} 35 & 35 & 20 \\ 42 & 42 & 27 \\ 19 & 19 & 12 \end{bmatrix}$$

$$= 35 + 35 + 20 + 42 + 0 + 27 + 19 + 19 + 12$$

$$= 458$$

Second Row

$$P_{21} = \begin{bmatrix} 29 & 46 & 44 \\ 16 & 25 & 21 \\ 66 & 120 & 116 \end{bmatrix}$$

$$= 29 + 46 + 44 + 16 + 0 + 21 + 66 + 120 + 116$$

$$= 458$$

$$C_{22} = \begin{bmatrix} 46 & 44 & 42 \\ 25 & 21 & 19 \\ 120 & 116 & 154 \end{bmatrix}$$

$$= 46 + 44 + 42 + 25 + 0 + 9 + 120 + 116 + 154$$

$$= 566$$

$$C_{23} = \begin{bmatrix} 44 & 42 & 42 \\ 21 & 19 & 19 \\ 116 & 154 & 114 \end{bmatrix}$$

$$= 44 + 42 + 42 + 21 + 0 + 19 + 116 + 154 + 114$$

$$= 472$$

Third Row

$$C_{31} = \begin{bmatrix} 16 & 25 & 21 \\ 66 & 120 & 116 \\ 74 & 116 & 174 \end{bmatrix}$$

$$= 16 + 25 + 21 + 66 + 0 + 116 + 74 + 116 + 174 \\ = 708$$

$$C_{32} = \begin{bmatrix} 25 & 21 & 19 \\ 120 & 116 & 154 \\ 216 & 174 & 254 \end{bmatrix}$$

$$= 25 + 21 + 19 + 120 + 0 + 154 + 216 + 174 + 254 \\ = 888$$

$$C_{33} = \begin{bmatrix} 21 & 19 & 19 \\ 116 & 154 & 114 \\ 174 & 252 & 172 \end{bmatrix}$$

$$= 21 + 19 + 19 + 116 + 0 + 114 + 174 + 252 + 172$$

$$= 887$$

$$C_{34} = \begin{bmatrix} 19 & 19 & 12 \\ 154 & 114 & 62 \\ 252 & 172 & 112 \end{bmatrix}$$

$$= 19 + 19 + 12 + 154 + 0 + 62 + 252 + 172 + 112$$

$$= 802$$

Fourth Row

$$C_{41} = \begin{bmatrix} 66 \\ 74 \\ 70 \end{bmatrix}$$

$$= 66 + 120 + 116 \\ = 1000$$

$$C_{42} = \begin{bmatrix} 120 \\ 216 \\ 210 \end{bmatrix}$$

$$= 120 + 116 \\ = 1488$$

$$C_{43} = \begin{bmatrix} 116 \\ 174 \\ 120 \end{bmatrix}$$

$$= 116 + 154 \\ = 1320$$

$$C_{44} = \begin{bmatrix} 154 \\ 252 \\ 112 \end{bmatrix}$$

$$= 154$$

$$= 122$$

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Fourth row

$$C_{41} = \begin{bmatrix} 66 & 120 & 116 \\ 74 & 216 & 174 \\ 70 & 210 & 170 \end{bmatrix}$$

$$= 66 + 120 + 116 + 74 + 0 + 174 + 70 + 210 + 170 \\ = 1000$$

$$C_{42} = \begin{bmatrix} 120 & 116 & 154 \\ 216 & 174 & 252 \\ 210 & 170 & 250 \end{bmatrix}$$

$$= 120 + 116 + 154 + 216 + 0 + 252 + 210 + 170 + 250 \\ = 1488$$

$$C_{43} = \begin{bmatrix} 116 & 154 & 114 \\ 174 & 252 & 172 \\ 170 & 250 & 170 \end{bmatrix}$$

$$= 116 + 154 + 114 + 174 + 0 + 172 + 170 + 250 + 170 \\ = 1320$$

$$C_{44} = \begin{bmatrix} 154 & 114 & 62 \\ 252 & 172 & 112 \\ 250 & 170 & 110 \end{bmatrix}$$

$$= 154 + 114 + 62 + 252 + 0 + 112 + 250 + 170 + 110 \\ = 1224$$

Output

$$\begin{bmatrix} 225 & 258 & 250 & 209 \\ 458 & 566 & 887 & 472 \\ 708 & 981 & 887 & 802 \\ 7000 & 1488 & 1320 & 1224 \end{bmatrix}$$

Step 2

Window shape - [2, 2]

Stride - 2

Convolutional Output

$$\begin{bmatrix} 225 & 258 & 250 & 209 \\ 458 & 566 & 887 & 472 \\ 708 & 981 & 887 & 802 \\ 1000 & 1488 & 1320 & 1224 \end{bmatrix}$$

Pooling block: 1

$$\begin{bmatrix} 225 & 258 \\ 458 & 209 \end{bmatrix} = 566$$

Block 2

$$\begin{bmatrix} 250 & 209 \\ 552 & 472 \end{bmatrix} = 552$$

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$$\text{Block 3} \quad \begin{bmatrix} 708 & 981 \\ 1000 & 1420 \end{bmatrix} = 1488$$

Block 4

$$\begin{bmatrix} 887 & 802 \\ 1320 & 1224 \end{bmatrix} = 1320$$

After max pooling

$$M = \begin{bmatrix} 566 & 552 \\ 1488 & 1320 \end{bmatrix}$$

4) Given,

$$X = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

$$\text{Filter} = 2 \\ \text{Stride} = 2$$

$$\text{Output size} = \frac{N-F}{S} + 1$$

$$= \frac{4-2}{2} + 1 = 2$$

Apply average pooling

$$\text{Block 1} \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}$$

$$= \frac{1 + 2 + 5 + 6}{4}$$

$$= 3.5$$

$$\text{Block 2} \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix}$$

$$= \frac{3 + 4 + 7 + 8}{4}$$

$$= 5.5$$

$$= 5.5$$

Block 4

$$\begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix}$$

$$= \frac{11 + 12 + 15 + 16}{4}$$

$$= 13.5$$

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Final Answer

$$M_{avg} = \begin{bmatrix} 3-5 & 5-5 \\ 11-5 & 13-5 \end{bmatrix}$$

~~NO~~